



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Mid-term Exam: : Trimester: Spring 2026

Course Code: CSE 1111, Course Title: STRUCTURED PROGRAMMING LANGUAGE

Time: 1 hour 30 min Total Marks: 30

(Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.)

Answer all the Questions. Numerical values in the right margin indicate full marks.

- 1 a) Show the **manual tracing** of the following code 3

```
#include<stdio.h>
int main() {
int sum=24, i=23;
    i=i%4;
    sum=sum/i++;
    printf("sum=%5d i=%d\n", --sum, ++i);
    sum*=i--;
    printf("sum=%5d i=%d", sum++, i--);
    return 0;
}
```

- b) Find **output** of the following code segment. Remember that manual tracing is not needed. 3

```
int num=25, sum=15;
printf("%4d%3d\n", num, sum);
switch((num+sum)/3+1){
    case 2: printf("%d %d\n", num+2, sum);

    case 5: sum=sum+num;
            printf(" %d%d\n", num+2, sum);
    case 6:
            printf("%d\n", sum+5); break;
    default:
            printf("%d\n", num+10);
}
printf("\n\n%d %d", num, sum);
```

- 2 a) **Rewrite** the code segment using “switch” without changing the logical meaning 4

```
int a, b, c;
scanf("%d%d%d", &a, &b, &c);
if ( (a+b-c== 0) || (a+b-c==4)) {
    printf("a = %d\n", a+b);
}
else if ((a+b-c) < 5) {
    printf("b = %d\n", a-b);
}
else
    printf("c = %d\n", c);
```

- b) **Manually trace** the following code segment 4

```
int num=964, i=10;
printf("%d %d\n", num, i);
while(num>100){
    num=num%i;
    printf("%d %d\n", num, i)
    i=i*10;
}
printf("%d %d\n", num, i);
```

- 3 a) Draw a **flow chart** to evaluate the following series 4
 $2 - 4 + 6 - 8 + \dots - 100$

- b) Write a **program** that prints the summation of square of a digit at all even positions of an input integer number from the right most digit. 4
 For example, num=28396 and Sum=0

Digit	Square	Sum
6	-	-
9	9^2	$0+81=81$
3	-	-
8	8^2	$81+64=145$
2	-	-

- 4 a) Write a **program** that stores input integer data into an array A of size n, and find the moving average of window size 3 and store it into a floating-point array B. Finally, the program prints all the elements in the array B. 4

For example, n=5

Input Array A: 10 33 67 56 45

New Array B: $(10+33+67)/3$ $(33++67+56)/3$ $(67+56+45)/3$

Output data Array B: 36.67 52.00 56.00

- b) **Manually trace** the following code segment 4

```
int i, j;
int A[3][4]={12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1};
for(i=0; i<3; i++){
    for(j=0; j<4; j++){
        if (i==j){
            printf("%d ", A[i][j]);
        }
    }
    printf("\n");
}
```